

Blackbody: Radiation from matter which has come into thermal eq. at a given temp  $T$ .

- $\rho(\nu, T) d\nu$ : energy per volume at freq. between  $\nu$  and  $\nu + d\nu$

Assumed radiation field was a sum of  $e^{i\vec{q} \cdot \vec{x}}$  with  $\vec{q} = 2\pi\vec{n}/L$

$$\vec{n} = (n_1, n_2, n_3) \quad n_i \in \mathbb{Z}$$

$$\lambda = \frac{2\pi}{|\vec{q}|}, \quad q: \text{wave number}$$

$$c = \lambda \nu \Rightarrow \nu = c/\lambda = \pi c / 2\pi$$

$$N = \frac{|\vec{n}|^2}{L}$$

$N(\nu) d\nu$ : # of normal modes between  $\nu$  and  $\nu + d\nu$

$$N(\nu) d\nu = 2 \left( 4\pi |\vec{n}|^2 d|\vec{n}| \right) = 8\pi \left( \frac{L}{c} \right)^3 \nu^2 d\nu$$



$$E \sim k_B T$$

$$\rho(\nu, T) d\nu =$$

$$\frac{\langle E(\nu) \rangle}{1^3} N(\nu) d\nu = \frac{8\pi k_B T \nu^2 d\nu}{c^3}$$

Planck:  $p(N, T) dN = \frac{8\pi h}{c^3} \frac{N^3 dN}{\exp(hN/k_B T) - 1}$   
 Agrees with classical approach for  $N/T \ll \frac{k_B}{h}$

$$\int p(N, T) dN = G_B T^4, \quad G_B = \frac{8\pi^5 k_B^4}{15 h^3 c^3}$$

\* Photoelectric effect:

→ Minimum:  $h\nu_{\min}$

emitted photon will have energy

$$E_{\text{emitted}} = h(\nu - \nu_{\min})$$



$$E_{\text{in}} = E_{\text{out}} + E_{\min}$$

$$h\nu - E_{\min} = F_{\text{out}}$$

I see, so  $F_{\text{out}} = h\nu - h\nu_{\min}, \quad \nu \geq \nu_{\min}$

→  $\langle E \rangle = k_B T$  using classical

$$\langle E \rangle \text{ for } \propto h\nu \Rightarrow h\nu / (e - 1)$$

α-particles:  $\text{He}^4$  nuclei:  $2P + 2N$

$$E_{\min} = h\nu_{\min} \rightarrow \text{Binding energy.}$$



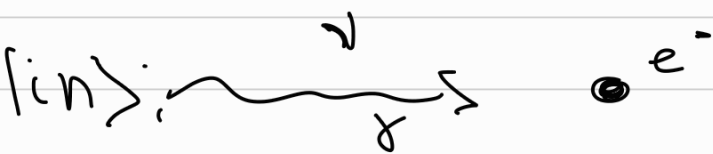
$e^-$ : free particle

$$E = h\nu \Rightarrow p = h\nu/c$$

Since  $E^2 = (m_e c^2)^2 + (pc)^2 = (h\nu)^2$

\* In order that  $m_e^2 c^4 = 0$

\*  $h\nu = pc$  \*



$$E_{in} = h\nu + m_e c^2$$

$$E_{out} = h\nu' + \sqrt{(m_e c^2)^2 + (pc)^2}$$

$$p_{in} = \frac{h\nu}{c}$$

$$p_{out} = -h\nu'/c + p_e$$

$$\therefore p_e = \frac{h}{c^2} (\nu + \nu')$$

$\therefore$

$$h\nu + m_e c^2 = h\nu' + \sqrt{(m_e c^2)^2 + \frac{h^2}{c^2} (\nu + \nu')^2}$$

$$(h(\nu - \nu') + m_e c^2)^2 = m_e^2 c^4 + h^2 (\nu + \nu')^2$$

$$h^2 (\nu - \nu')^2 + 2h(\nu - \nu')m_e c^2 + m_e^2 c^4 = m_e^2 c^4 + h^2 (\nu + \nu')^2$$

$$\therefore h^2 (\nu - \nu')^2 + 2h(\nu - \nu')m_e c^2 = h^2 (\nu + \nu')^2$$

$$h^2 \nu^2 - 2h^2 \nu \nu' + h^2 \nu'^2 + 2h(\nu - \nu')m_e c^2$$

$$= h^2 \nu^2 + 2\nu \nu' h^2 + h^2 \nu'^2$$

$$4h^2 \nu \nu' = 2h(\nu - \nu')m_e c^2$$

$$\nu' [4h\nu + 2m_e c^2] = 2\nu m_e c^2$$

$$\nu' = \frac{\nu m_e c^2}{2h\nu + m_e c^2}$$

Using  $\lambda' = \frac{c}{\nu'}$  ;  $\lambda' = \lambda + \frac{2h}{m_e c}$